Group projects Programming in Python II

Deadline: 10/12/2023, 23:59, please submit your *.py or *.ipynb files using the designated folder on the K2 site of the course.

Project 1: Team 5 (PROSPERI Armand,ORSAZ Léna,GRIFFOUL Victorine), Team 1 (MICHEL Bastie, OBRAN James, MENDY Gélaze, POIRIER Thibault)

The task is to predict the logarithm of the hourly wages (in USD) (variable *lwage*) based on education (variable *educ*), work experience (variable *exper*), tenure (variable *tenure*) and gender (variable $female$). The gender is encoded in the variable $female$ which is 1 for females, zero for males. The .csv file can be found at

`https://kaggle.com/datasets/02f69ffd32ee24c19c78526e7dee71cfe7 5a475a78c48b66c63fc11dc1d09e75`.

Use various models (linear regression, SVM, decision tress, neural networks) for predicting the logarithm of wage. Compare the various models in terms of accuracy of prediction. Which one is the best ? Select one or two independent variables (say only the tenure and gender, or education and tenure, etc.) and use them to build models for predicting the logarith of wage. Which variables are the most give the best model for predicting wage ? Repeat the exercise separately for males and females and for the wage instead of its logarithm.

Project 2: Team 2 (RABEMANANTSOA Mirindra, MOREAU Alice, LEFEBVRE Simon), Team 3 (FIRMANSYAH Freddy, YAMIN Muhammad, ZAFFARONI Tommaso, ZAFFAGNI Ruben)

The task is to predict the logarithm of the house prices (variable *lprice*) using the following variables: logarithm of the lot size (variable *llotisize*), logarithm of the surface of the house (variable *lsqrtft*), number of bedrooms (variable *bdrms*). The .csv file can be found at https://kaggle.com/datasets/eb36254d39a218a99ac169cedb764d10b6468df9aff710c1e3508651170ff355.

Use various models (linear regression, SVM, decision tress, neural networks) for predicting the logarithm house prices. Compare the various models in terms of accuracy of prediction. Which one is the best ? Select one or two independent variables and use them to build models for predicting the logarithm of houses prices. Which variables give the best model for predicting the logarithm of house prices ? Repeat the exercises with the house prices instead of their logarithm.

Project 3: Team 4 (Christodoulou Georges, Sebbah Yohan Simon, Birbes Eva)) Team 6 (LALEYE Nadirath, Nagori Mehul, Sharma Supriya, Yuan Gong, KELLER Annemarie)

The task is to predict house prices in different communities based on air pollution level (represented by the concentration of nitrogine oxid), distance from employment centers, average student-teacher ratio in local schools and average number of rooms, and crime rate The variables are as follows:

- *price*
  median housing price
- *crime*
  crimes committed per capita
- *nox*
  nitrogine oxide, parts per 100 mill.
- *rooms*
  avg number of rooms per house
- *dist*
  weighted dist. to 5 employ centers
- *stratio*
  average student-teacher ratio
- *lprice*
  $\log(price)$
- *lnox*
  $\log(nox)$

The .csv file can be found at
https://kaggle.com/datasets/f06415afa98ac5a0a50ad7fdec3c4b526023dcf2a1b39435e6f2b6a43eddf88e.

Use various models (linear regression, SVM, decision tress, neural networks) for predicting house prices. Compare the various models in terms of accuracy of prediction. Which one is the best ? Select one or two independent variables and use them to build models for predicting houses prices. Which variables give the best model for predicting house prices ? Try to predict the logarithm of house prices instead of house prices. Do you see any improvement in performance of the models ?

Project 4: Team 7 (ZHANG Fengyuan, CHEN Tianle, ZHANG Gening, LI WEI, ZHANG Zhiling), Team 8 (LIN Zhan, BENLAHCEN Yousra, BENHAMOU Ethan, Tian TONG))

The task is to predict the price growth based on the monthly growth of the hourly wage. The variables are as follows

- *price*
  consumer price index
- *gwage*
  increase in hourly wage in the current month
- $gwage_i$
  $i = 1, \ldots, 12$ increase in hourly wage in month = the current month$-i$

The .csv file can be found at `https://www.kaggle.com/datasets/mihalypetreczky/wageprcdataset`.

Use various models (linear regression, SVM, decision tress, neural networks) for predicting price growth. Compare the various models in terms of accuracy of prediction. Which one is the best ? Adapt the models to predict the price *price* based on $gwage, gwage_1, \ldots, gwage_n$, for $n = 1, 2, \ldots, 12$. Which variables give the best model for predicting price growth ? Try to predict the logarithm of house prices instead of house prices. Do you see any improvement in performance of the models ?

Project 5: Team 9 (FLETCHER SAMUEL,EICHER Clarine, HAGGREN Ilari, PALEARI Beatrice)

The task is to predict the inventory change based on GDP growth and on real interest rates. More precisely, the variables are as follows:

- *cinven*
  annual growth of inventory

- *cgdp*
  annual growth of GDP
- $r3$
  real interest rate
- $cr3$
  annual growth of the real interest rate
- *gdp*
  annual gdp

The .csv file can be found at `https://www.kaggle.com/datasets/mihalypetreczky/invendataset`.

Use various models (linear regression, SVM, decision tress, neural networks) for predicting inventory. Compare the various models in terms of accuracy of prediction. Which one is the best ? Select one or two independent variables and use them to build models for predicting inventory. Which variables give the best model for predicting inventory ?

Project 6:Team 11(Muhammad YAMIN, Ruben ZAFFAGNI, Tommaso ZAFFARONI), Team 12 (Fengyuan ZHANG, Gening ZHANG, Zhiling ZHANG)

The task to predict logarithm of wages based on education, experience and IQ, hours worked. The variables are as follows:

- *lwage*
  logarithm monthly earnings
- *hours*
  average weekly hours
- *IQ*
  IQ score
- *educ*
  years of education
- *exper*
  years of work experience

The .csv file can be found at
`https://kaggle.com/datasets/053a610ff5d8f491e3ee24c7eadc563cdd094ff395e72ba6734ebeec4ab1ca59`.

Use various models (linear regression, SVM, decision tress, neural networks) for predicting the logarithm of wage. Compare the various models in terms of accuracy of prediction. Which one is the best ? Select one or two independent variables and use them to build models for predicting the logarithm of wage. Which variables give the best model for predicting the logarithm of wage ? Repeat the exercise for wage instead of its logarithm. Do you see any difference in the performance of various models ?